

TMA4315 Exercise 1: Linear models for Gaussian data

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Part 1: Explanatory analysis of the dataset

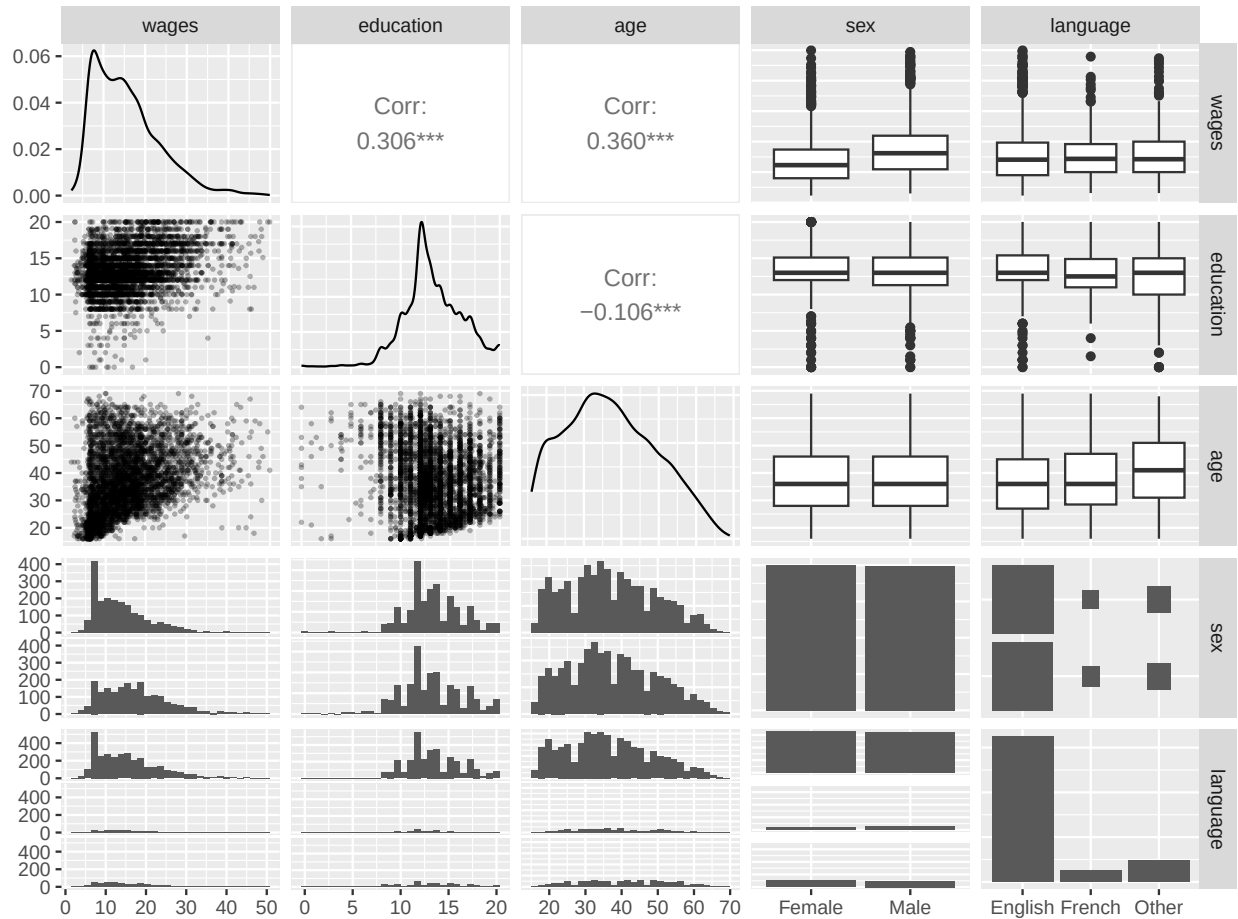
First of all we import the dataset and we prepare the data for the analysis.

```
library(car)
data(SLID, package = "carData")
SLID <- SLID[complete.cases(SLID), ]
```

After that we have 3987 observations and 5 variables: **wages** (the hourly wage rate), **education** (the number of years of schooling), **age** (in years), and the two categorical variables **sex** and **language**.

Then we plot a matrix of diagnostic plots for all the variables.

```
ggpairs(SLID, lower = list(continuous = wrap("points", alpha = 0.3, size = 0.5)))
```



On this plot we can see some interesting relations between the pairs of variables:

- **education** and **wages** have a moderate positive correlation of 0.306, and it is highly significant. It is the same thing for **age** and **wages**, with a correlation of 0.360. The two correlations are quite small, so we should not expect a very big R^2 later: a correlation of 0.306 gives only about $0.306^2 \approx 9\%$ of the variability of **wages**.
- the median wage is higher for men (16.20) than for women (12.29), but it is almost the same for the different languages (14.09, 14.34, 14.24). This means that the gender of a person can be strongly related to the wage, while the language has probably a much weaker effect. In the two cases we can also see that most of the wages are concentrated on low values, except a few outliers which are much higher. We can confirm this with the density plot of **wages**, which is skewed to the right.
- **age** and **education** have a very weak negative correlation of -0.106, but it is highly significant. This means that the two variables are almost uncorrelated, but not exactly. This is why the simple and the multiple regression estimates will not be the same in Part 3.
- the median education is the same for men and for women (13 years), and very close for the different languages (13, 12.5, 13). Thus we can say that education is not affected by these two variables. In the two cases the median education is quite high, except a few outliers with a much lower value.
- we can say the same thing for the **age** variable, but we can see that the median age of the people who speak an **other** language is higher (41 against 36).
- there is nothing special about the relation between age and language. But we can see that the split between men and women is almost equal (2001 against 1986), while the **language** variable is dominated by **English** (3244 of 3987 observations).

We need the following assumptions on the data if we want to do a multiple linear regression analysis. With the notations of the module 2 notes, the model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ is a **classical linear model** if:

1. $E(\boldsymbol{\varepsilon}) = \mathbf{0}$, so the relation between the response and the covariates is linear and additive:

$$y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik} + \varepsilon_i, \quad i = 1, \dots, n.$$

2. $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$. This means that the errors are homoscedastic (they all have the same variance σ^2 , which does not depend on the covariates) and that they are uncorrelated, $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$ for $i \neq j$.
3. The design matrix has full rank, $\text{rank}(\mathbf{X}) = k + 1 = p$. Without this we cannot invert $\mathbf{X}^T \mathbf{X}$ and $\hat{\boldsymbol{\beta}}$ is not unique.

We get the classical **normal** linear model if we add

4. $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I})$.

If we write it in the GLM way, the same model says that the pairs $(Y_i, \mathbf{x}_i)$ are independent with $Y_i \sim N(\mu_i, \sigma^2)$ (random component), $\eta_i = \mathbf{x}_i^T \boldsymbol{\beta}$ (systematic component) and the identity link $\mu_i = \eta_i$.

We only need assumption 4 for the exact t - and F -tests. As **wages** is clearly skewed to the right, it is the assumption that we trust the least here, and this is one more reason to use the asymptotic z - and χ^2 -tests in Part 2.

Part 2: Linear regression with the mylm package

Statistical Formulas and Theory

The `mym` package implements ordinary least squares (OLS) for a linear model $Y = X\beta + \varepsilon$. The functions in the package are built upon the following formulas:

- **Coefficient estimates:** $\hat{\beta} = (X^\top X)^{-1} X^\top Y$
- **Fitted values:** $\hat{Y} = X\hat{\beta}$
- **Residuals:** $e = Y - \hat{Y}$
- **Residual Sum of Squares (SSE):** $SSE = e^\top e$
- **Degrees of Freedom (df):** $df = n - p$ (where p is the number of parameters).
- **Total Sum of Squares (SST):** $SST = \sum (y_i - \bar{y})^2$
- **Covariance matrix:** $\widehat{\text{Cov}}(\hat{\beta}) = \hat{\sigma}^2 (X^\top X)^{-1}$, with $\hat{\sigma}^2 = \frac{SSE}{n-p}$
- **Standard Errors (SE):** $\text{SE}(\hat{\beta}_j) = \sqrt{\widehat{\text{Cov}}(\hat{\beta})_{jj}}$
- **Asymptotic z-test:** $z_j = \frac{\hat{\beta}_j}{\text{SE}(\hat{\beta}_j)}$ with p-value $2\Phi(-|z_j|)$.
- **Asymptotic χ^2 -test:** $\chi^2 = \frac{SSE_{\text{reduced}} - SSE_{\text{full}}}{\hat{\sigma}_{\text{full}}^2}$
- **Coefficient of determination:** $R^2 = 1 - \frac{SSE}{SST}$

a) & b) Fitting the model and calculating standard errors

We fit a simple linear regression model to the data with `wages` as the response and `education` as the predictor.

```
modell1 <- mylm(wages ~ education, data = SLID)
modell1b <- lm(wages ~ education, data = SLID)

print.mylm(modell1)
```

```
##
## Call:
## mylm(formula = wages ~ education, data = SLID)
##
## Coefficients:
## (Intercept)    education
##    4.9716912    0.7923091
```

```
print(modell1b)
```

```
##
## Call:
## lm(formula = wages ~ education, data = SLID)
##
## Coefficients:
## (Intercept)    education
##    4.9717    0.7923
```

The output confirms that our `mym` function yields the exact same coefficient estimates as the standard `lm` function.

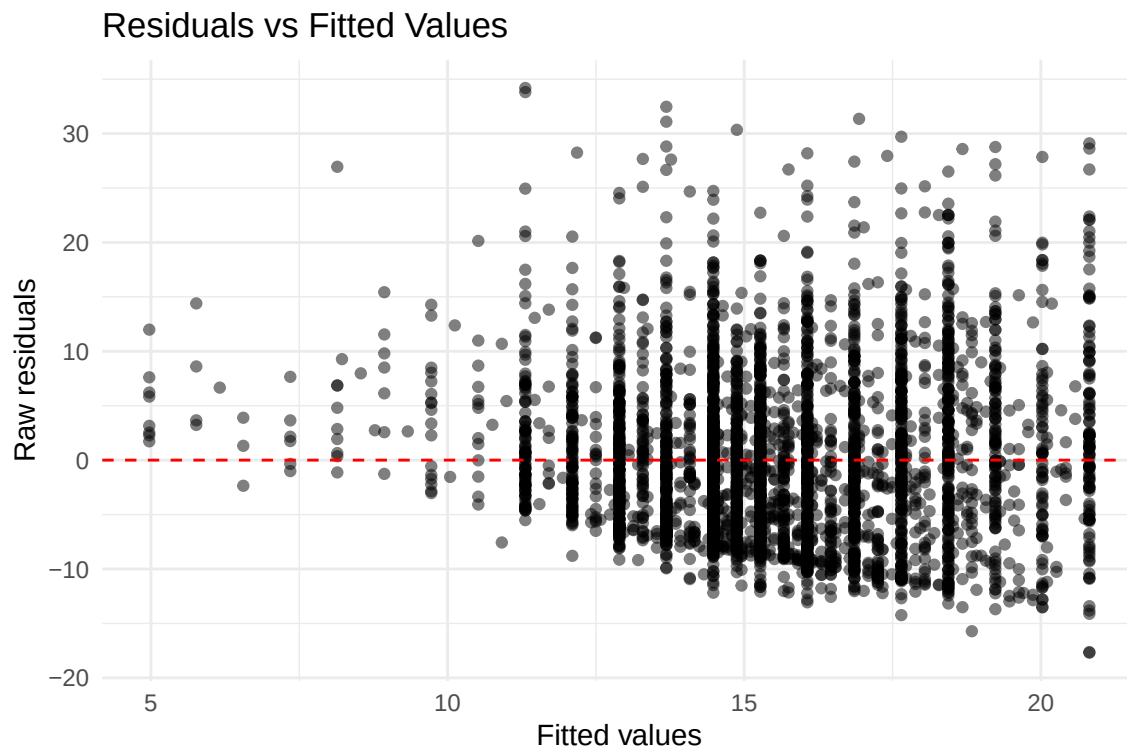
```
summary.mylm(model1)
```

```
## Summary of object
##
## Call:
## mylm(formula = wages ~ education, data = SLID)
##
## Coefficients:
##             Estimate Std. Error z value Pr(>|z|)
## (Intercept)  4.97169    0.53429   9.3052 < 2.2e-16 ***
## education    0.79231    0.03906  20.2842 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Multiple R-squared:  0.093586
```

Interpretation of parameters: * **Intercept** ($\hat{\beta}_0 \approx 4.97$): The expected hourly wage for a person with 0 years of education. This is extrapolated outside the data range and holds no practical meaning since wages cannot be negative. * **Education** ($\hat{\beta}_1 \approx 0.79$): For each additional year of education, the expected hourly wage increases by 0.79 units. * **Significance (z-test)**: The z-value for education is 20.28, yielding a p-value strictly close to 0. We confidently reject the null hypothesis ($H_0 : \beta_1 = 0$); education is a highly significant predictor.

c) Residuals plot

```
plot.mylm(model1)
```



Comment on the plot: The residuals versus fitted values plot shows a funnel shape, meaning the variance of the residuals increases as the fitted values increase. This violates the assumption of homoscedasticity.

d) ANOVA and Chi-squared test

```
results_anova <- anova.mylm(model1)

## Analysis of Variance Table
## Response: wages
##           Df Sum sq X2 value Pr(>X2)
## education 1  23096.20 411.45  1.7747e-91
```

Relationship between tests: In simple linear regression, the χ^2 statistic with 1 degree of freedom is equal to the square of the z -statistic for the slope coefficient. From our summary, $z \approx 20.284$, and $z^2 \approx 411.45$, matching the χ^2 value in the ANOVA table. The critical value for a 5% significance level is $z_{0.025} \approx 1.96$ and $\chi^2_{1,0.05} \approx 3.84$.

e) Coefficient of determination (R^2)

The coefficient of determination R^2 is **0.0936**.

Interpretation: Approximately **9.36%** of the total variability in hourly wages is explained by the variance in years of education. The remaining **90.64%** is unexplained by this model.

Part 3: Multiple linear regression

a) Fitting the multiple regression model

Because our `mylm` and `print.mylm` functions were strictly implemented using generalized matrix operations (specifically $\hat{\beta} = (X^\top X)^{-1} X^\top Y$), they automatically adapt to any number of predictors. No changes to the underlying R functions are necessary to handle multiple linear regression.

```
model2 <- mylm(wages ~ education + age, data = SLID)
print.mylm(model2)
```

```
##
## Call:
## mylm(formula = wages ~ education + age, data = SLID)
##
## Coefficients:
## (Intercept)  education          age
##  -6.0216529   0.9014644   0.2570898
```

b) Estimates, standard errors, and z-test

Similarly, our `summary.mylm` function uses the generalized covariance matrix formula $\hat{\sigma}^2(X^\top X)^{-1}$. Extracting the diagonal elements for the standard errors works flawlessly regardless of the model's dimensions.

```
summary.mylm(model2)
```

```
## Summary of object
##
## Call:
## mylm(formula = wages ~ education + age, data = SLID)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -6.0216529  0.6189238 -9.7292 < 2.2e-16 ***
## education    0.9014644  0.0357598 25.2089 < 2.2e-16 ***
## age          0.2570898  0.0089512 28.7212 < 2.2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Multiple R-squared:  0.249070
```

Interpretation of parameters: * **Intercept** ($\hat{\beta}_0 \approx -6.02$): The expected hourly wage for a theoretical person with 0 years of education and an age of 0. Even though this doesn't make much sense practically, it is consistent with the regression model. * **Education** ($\hat{\beta}_1 \approx 0.90$): This means that each additional year of education increases the hourly wage by 0.9, for equal age. * **Age** ($\hat{\beta}_2 \approx 0.26$): This means that each additional year of age increases the hourly wage by 0.26, for equal education. Thus, we can see that education is valued more highly than age in determining wages. * **Significance:** We can see from the summary that the p-values for age and education are far smaller than 5% (2.2e-16 for both), which means that age and education are highly significant predictors of wages.

c) Simple vs. Multiple Regression Estimates

The estimated parameter for **education** is not the same in the simple regression and in the multiple regression, because of the **omitted variable bias**. In the simple model, the coefficient of **education** has to take both its own effect on wages and the effect of the variable that we forgot (**age**). This happens because **age** and **education** are correlated in our dataset.

In the simple linear regression, the **education** coefficient absorbs both its direct effect on wages and the indirect effect of the omitted variable (**age**), since **age** and **education** are correlated (as seen in Part 1, $r = -0.106$). In the multiple linear regression, the coefficient isolates the **partial effect** of **education** on wages while explicitly controlling for **age**.

We can show these differences by running the individual models with **mylm** and comparing their coefficients:

```
# Fit the individual simple regressions
mod_edu <- mylm(wages ~ education, data = SLID)
mod_age <- mylm(wages ~ age, data = SLID)

# Display the isolated vs partial effects
cat("Simple regression education effect:  ", mod_edu$coefficients["education"], "\n")
```

```
## Simple regression education effect:      0.7923091
```

```
cat("Simple regression age effect:         ", mod_age$coefficients["age"], "\n\n")
```

```
## Simple regression age effect:            0.2331079
```

```
cat("Multiple regression education effect: ", model2$coefficients["education"], "\n")
```

```
## Multiple regression education effect: 0.9014644
```

```
cat("Multiple regression age effect:      ", model2$coefficients["age"], "\n")
```

```
## Multiple regression age effect:      0.2570898
```

```
““
```